

Pipeline Diagram: Input → Detect Source → Parse → Normalize → Merge → Confidence → Projection → Validation → JSON Output

1. System Architecture & Pipeline

- Built on a decoupled pipeline pattern executing sequentially:
- **Ingestion:** Identifies and validates unstructured (PDF) and structured (CSV) source file paths.
 - **Parsing & Normalization:** Extracts fields and formats them to target standards prior to consolidation.
 - **Consolidation:** Executes conflict resolution policies and computes lineage metadata.
 - **Presentation:** Customizes schema projection shapes and formats values for validation.

2. Canonical Schema Design

- Defines the single source of truth for candidate attributes via strict Pydantic schemas:
- **candidate_id:** Unique deterministic string generated per candidate.
 - **location:** Structured sub-schema containing *city*, *region*, and *country*.
 - **links:** Categorized endpoints mapping *linkedin*, *github*, *portfolio*, and *other* URL lists.
 - **history:** Structured arrays for *experience* and *education* chronologically ordered.
 - **metadata:** Lists tracking *provenance* logs and *field_confidence* levels.

3. Normalization Strategy

- Standardizes attributes individually before merging to ensure deterministic outcomes:
- **Phones:** Formatted to **E.164** using the Google *phonenumbers* library. Mock numbers fall back to possible structures to ensure robust handling.
 - **Dates:** Standardized to **YYYY-MM**. Relies on *python-dateutil* to parse free-text inputs, resolving current timelines to 'Present'.
 - **Country:** Maps inputs to **ISO-3166-1 alpha-2**. Translates common US states (e.g. 'CA' → 'US') and defaults unresolved fields.
 - **Skills:** Resolves skills (e.g., 'cpp', 'C Plus Plus' → 'C++') using exact synonym mappings and fuzzy matching ('rapidfuzz').

4. Merge & Conflict Resolution Strategy

- Combines multiple candidate sources using a priority-based precedence logic:
- **Precedence hierarchy:** Resume PDF (Priority 2) > Recruiter CSV (Priority 1) > Unknown (0).
 - **Conflict Policy:** Single-value fields are resolved by taking the value from the highest priority source.
 - **Array Unioning:** Deduplicates lists (emails, phones) and aggregates skills.
 - **Timeline Alignment:** Overlapping experiences (same company & start year) are consolidated and sorted by start date.

5. Explainable Confidence Scoring

- Assigns explicit confidences representing the quality and certainty of the data:
- **Resume (Direct):** 0.95 confidence weight.
 - **CSV (Direct):** 0.90 confidence weight.
 - **Regex extraction:** 0.80 confidence weight.
 - **Heuristics fallback:** 0.60 confidence weight.
 - **Unknown:** 0.30 fallback weight.
 - **Overall score:** Mathematical average of populated field-level confidence ratings.

6. Provenance Model

- Ensures full audit lineage by tracking extraction lineage for every field:
- Stores **field** name, **source** filename, and extraction **method**.
 - Lineage is recorded dynamically for all values, tracking the winning source for resolved conflicts.

7. Configurable Runtime Projection Layer

- Runtime Config:** Supports field inclusion, field renaming, normalization rules, provenance/confidence toggles, and configurable missing-value behavior without code changes.
- **Field Selection:** Restructure output using `include_fields` list.
 - **Field Aliasing:** Rename fields dynamically (e.g. `full_name` → `name`).
 - **Metadata Filtering:** Enables toggling metadata lists (confidence, provenance).
 - **Missing Fields:** Handles missing data by outputting 'null' values or 'exclude' (removing empty keys).

8. Schema Validation & Edge Cases

- **Validation:** Standardizes inputs using Pydantic. Reports syntax and formatting errors as readable messages.
- **Edge cases:** Gracefully handles empty source files, missing credentials, duplicate candidate records, and malformed files.

9. Assumptions & Future Enhancements

- **Assumptions:** Raw text is extractable from PDFs. Candidates sharing primary emails represent the same entity.
- **Future Enhancements:** Support additional parsers (LinkedIn, GitHub JSON, ATS XML). Integrate OCR (Tesseract) for scanned resumes.